

0. Introduction

Young-geun Kim

Department of Mathematical Sciences

MTH345: Deep Generative Models: A Statistical Perspective



ULSAN NATIONAL INSTITUTE OF
SCIENCE AND TECHNOLOGY

Outline

1. INTRODUCTION

2. COURSE MODULES & SCHEDULE

Motivating Example



- ▶ This course aims to cover the mathematical foundations of **generative models**, extending to early deep learning approaches.
- ▶ As a motivating example, we consider learning and generating outcomes from the unique die shown above.

Left: A replica at the Gyeongju National Museum. Right: Inscriptions on each face recorded at the time of excavation.

Motivating Example



- ▶ *Juryeonggu* (酒令具; AD668 ~ AD935) is an ancient die used for drinking games during a historical dynasty in what is now South Korea.
- ▶ It has 14 faces, six square ($\square$) and eight hexagonal ($\hexagon$), with each face assigning a penalty, e.g., 三盞一去: Drink three cups at once.
- ▶ **Q:** How can we model the stochastic nature of this die, and computationally generate simulated outcomes based on it?

Motivating Example

- ▶ *Probabilistic models* serve an essential role in describing the uncertainty of events arising from such random experiments.
- ▶ For example, we may introduce $\mathbf{p} = (p_1, \dots, p_{14})^T \in \{\mathbf{p} : p_i \geq 0, \sum_{i=1}^{14} p_i = 1\}$ to represent the probability of each face:

Outcome	1st 	2nd 	3rd 	4th 	5th 	6th 	1st 
Probability	p_1	p_2	p_3	p_4	p_5	p_6	p_7

Outcome	2nd 	3rd 	4th 	5th 	6th 	7th 	8th 
Probability	p_8	p_9	p_{10}	p_{11}	p_{12}	p_{13}	p_{14}

- ▶ With $\mathbf{p}$, the probability of observing any of square faces can be expressed as:

$$p_1 + \dots + p_6 \tag{1}$$

Motivating Example

- When $\mathbf{p} = (1/14, \dots, 1/14)^T$, it corresponds to the uniform model, where all faces are equally likely to occur:

Outcome	1st 	2nd 	3rd 	4th 	5th 	6th 	1st 
Probability	1/14	1/14	1/14	1/14	1/14	1/14	1/14

Outcome	2nd 	3rd 	4th 	5th 	6th 	7th 	8th 
Probability	1/14	1/14	1/14	1/14	1/14	1/14	1/14

- The probability of observing any of the square faces is:

$$p_1 + \dots + p_6 = \frac{6}{14}. \quad (2)$$

- In a similar manner, we can express the probability of any event using this probability table parameterized by $\mathbf{p}$.

Motivating Example

- When $\mathbf{p} = (\underbrace{1/12, \dots, 1/12}_6, \underbrace{1/16, \dots, 1/16}_8)^T$, it yields another probability table:

Outcome	1st $\square$	2nd $\square$	3rd $\square$	4th $\square$	5th $\square$	6th $\square$	1st $\hexagon$
Probability	1/12	1/12	1/12	1/12	1/12	1/12	1/16

Outcome	2nd $\hexagon$	3rd $\hexagon$	4th $\hexagon$	5th $\hexagon$	6th $\hexagon$	7th $\hexagon$	8th $\hexagon$
Probability	1/16	1/16	1/16	1/16	1/16	1/16	1/16

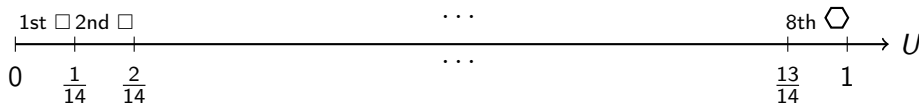
- The probability of observing any of the square faces is now:

$$p_1 + \dots + p_6 = \frac{1}{2}. \quad (3)$$

Motivating Example

- ▶ For a given $\mathbf{p}$, we can use the corresponding probabilistic model to generate synthetic samples.
- ▶ For example, when $\mathbf{p} = (1/14, \dots, 1/14)^T$, we can sample a number U from $[0, 1]$ randomly, and determine the outcome as follows:

$$\text{Outcome} = \begin{cases} \text{1st } \square & \text{if } 0 \leq U \leq 1/14 \\ \text{2nd } \square & \text{if } 1/14 < U \leq 2/14 \\ \vdots & \vdots \\ \text{8th } \hexagon & \text{if } 13/14 < U \leq 1 \end{cases}$$



Motivating Example

- ▶ Let's say that we sample the following 20 numbers:

$$\begin{aligned} &.3421, .7812, .0534, .9123, .4456, .6789, .2345, .8901, .5678, .1122, \\ &.9987, .4321, .7654, .0987, .3344, .8876, .5543, .2109, .6543, .1789 \end{aligned} \quad (4)$$

- ▶ Based on the previous rule, the corresponding synthetic samples are:

$$\begin{aligned} &5\text{th } \square, 5\text{th } \diamond, 1\text{st } \square, 7\text{th } \diamond, 1\text{st } \diamond, 4\text{th } \diamond, 4\text{th } \square, 7\text{th } \diamond, 2\text{nd } \diamond, 2\text{nd } \square, \\ &8\text{th } \diamond, 1\text{st } \diamond, 5\text{th } \diamond, 2\text{nd } \square, 5\text{th } \square, 7\text{th } \diamond, 2\text{nd } \diamond, 3\text{rd } \square, 4\text{th } \diamond, 3\text{rd } \square. \end{aligned} \quad (5)$$

Intuitively, as we repeat this process infinitely many times, the (empirical) probability table of these synthetic samples will converge to the one defined by $\mathbf{p}$.

- ▶ **Q:** Each $\mathbf{p}$ defines a distinct probability table, generating a different synthetic dataset. What would be the optimal choice of $\mathbf{p}$ to best mimic the real data?

Motivating Example

- ▶ The (*density*) *estimation* is the process of finding the optimal $\mathbf{p}$ based on *data*.
- ▶ Wang et al. (2005) conducted an experiment rolling the die $n = 7,000$ times using two different materials: wood and Fiberglass Reinforced Plastic (FRP).
- ▶ The observed relative frequencies for the wooden die are as follows:

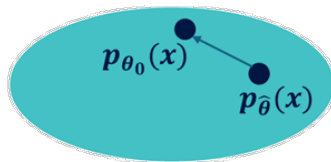
Outcome	1st 	2nd 	3rd 	4th 	5th 	6th 	1st 
Relative Frequency	$\frac{523}{7000}$	$\frac{533}{7000}$	$\frac{477}{7000}$	$\frac{525}{7000}$	$\frac{518}{7000}$	$\frac{492}{7000}$	$\frac{456}{7000}$

Outcome	2nd 	3rd 	4th 	5th 	6th 	7th 	8th 
Relative Frequency	$\frac{512}{7000}$	$\frac{506}{7000}$	$\frac{488}{7000}$	$\frac{505}{7000}$	$\frac{487}{7000}$	$\frac{482}{7000}$	$\frac{496}{7000}$

- ▶ This yields the empirical estimate $\hat{\mathbf{p}} = \left(\frac{523}{7000}, \dots, \frac{496}{7000} \right)^T$.

Motivating Example

Model Space
 $\{p_{\theta}(x) | \theta \in \Theta\}$



- ▶ We formalize the process illustrated in the example as follows.
- ▶ Let X be our target of interest (e.g., whether the die outcome is 1st $\square$ ($X = 1$) or not ($X = 0$)). We denote the parameter by θ (e.g., $\mathbf{p}$) and the probability by:

$$p_{\theta}(x) := P_{\theta}(X = x), \quad (6)$$

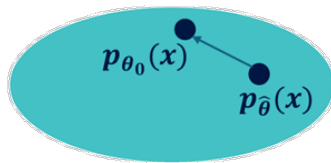
where P_{θ} represents the probability parameterized by $\mathbf{p}$.

- ▶ For example, the probabilities regarding the die example can be expressed as:

$$p_{\theta}(x = 1) = p_1 \quad \text{and} \quad p_{\theta}(x = 0) = p_2 + \cdots + p_{14} = 1 - p_1. \quad (7)$$

Motivating Example

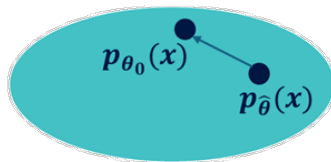
Model Space
 $\{p_{\theta}(x) | \theta \in \Theta\}$



- ▶ After specifying a class of probabilistic models, the model space, we then aim to find the optimal choice among them.
- ▶ In the die example, the model space can express all probability tables. Thus, it includes the true model $p_{\theta_0}(x)$, where θ_0 represents the unknown true parameter.
- ▶ Statistics justifies that the empirical estimator $\hat{\theta}$ converges to θ_0 as the sample size diverges.

Motivating Example

Model Space
 $\{p_{\theta}(x) | \theta \in \Theta\}$



- ▶ Q: What if the data are multidimensional? What if the true model is so complicated that the model space does not include the true model, say $p_{\text{data}}(x)$?
- ▶ Q: Furthermore, when it is impractical to employ the empirical estimator $\hat{\theta}$, how can we update $p_{\theta}(x)$ so that it converges to $p_{\text{data}}(x)$ in some sense?

Deep Generative Model



- ▶ The previous example illustrates the process of constructing probabilistic models, estimating the optimal models, and generating synthetic samples based on them.
- ▶ However, the high-dimensional and complex structures of recent data necessitate advanced statistical methods.
- ▶ Recent models have enabled the learning and generation of such complex data. They are now commonly referred to as *Generative AI* or *Deep Generative Models*.

Images are from the CelebA dataset (Liu et al., 2015).

Deep Generative Model



- ▶ Let $\mathbf{X} = (X_1, \dots, X_d)^T$ be an observation vector following $p(\mathbf{x})$, where, for example,

$$d = 1,024 \text{ (width)} \times 1,024 \text{ (height)} \times 3 \text{ (RGB color channels)} \approx 3.15\text{M} \quad (8)$$

for the color images shown above.

- ▶ Each pixel color value ranges from 0 to 255, comprising a total of 256 discrete integer values.

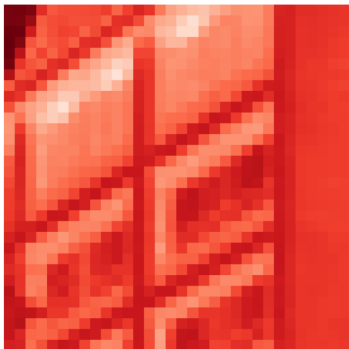
Deep Generative Model



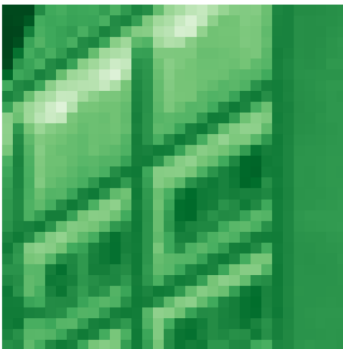
Source: Park (2022); Photograph by Kyung-chae Kim.

Deep Generative Model

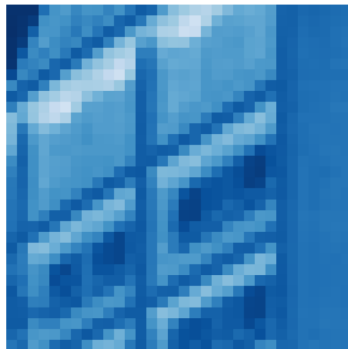
Red Channel



Green Channel



Blue Channel



Blue Channel

[illegible]

Deep Generative Model

- ▶ Deep generative models learn the *distribution* (e.g., probability table) of $\mathbf{X}$:

$$\begin{aligned} p_{\text{data}}(\mathbf{x}) &= P_{\text{data}}(\mathbf{X} = \mathbf{x}) \\ &= P_{\text{data}}(X_1 = x_1, \dots, X_d = x_d). \end{aligned} \tag{9}$$

- ▶ With $p_{\text{data}}(\mathbf{x})$, we can augment data, investigate cluster structures, and detect distinct patterns in data.

Application: Image Generation



- ▶ Generative models can synthesize photo-realistic data without prior information such as age and race in face generation.
- ▶ These models can be used to augment data, generate privacy-free samples, and enhance virtual reality experiences.

The generated image is from <http://thispersondoesnotexist.com/>

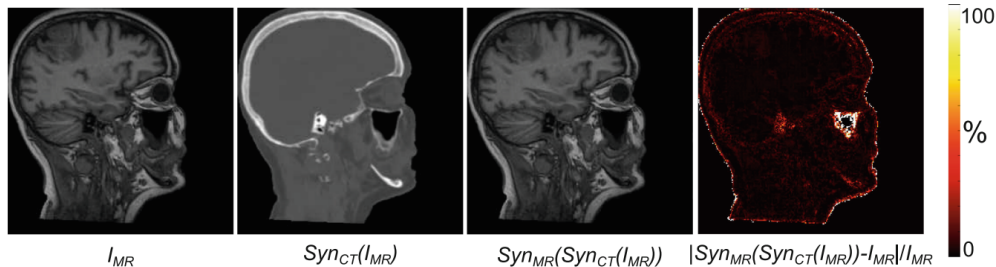
Application: Text-to-Image Generation



- ▶ The images are generated based on the following text description:
“There is a clean desk in the middle. Outside the window, a whale shark is swimming in the dark night sky above Manhattan.”
- ▶ The generated images reflect the semantic information provided in the text.

Images are generated with DALL-E-2 (Ramesh et al., 2022).

Application: Cross Modality Transfer



- ▶ Generative models are also useful in imputing missing modalities, e.g., magnetic resonance (MR) $\rightarrow$ computed tomography (CT), or enhancing data resolution.

Images are from Wolterink et al. (2017).

Outline

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Challenges

- ▶ For data such as 4K-resolution color images, the dimensionality is $d = 3,840 \times 2,160 \times 3 \approx 24\text{M}$.
- ▶ When we introduce parameters for every possible configuration of pixel values, e.g.,

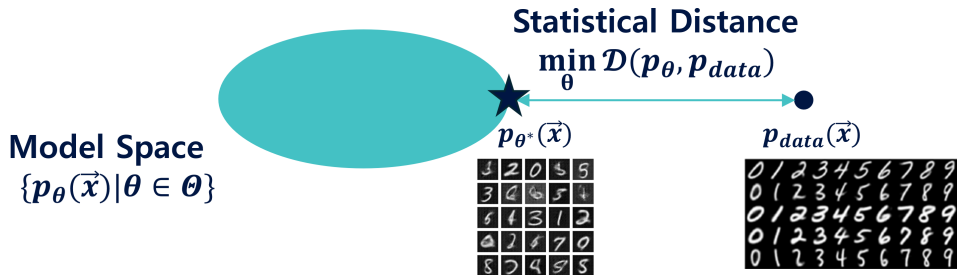
$$p_{0,0,\dots,0} := P(X_1 = 0, X_2 = 0, \dots, X_d = 0), \quad (10)$$

we can express any possible probability table. However, the number of parameters is astronomically large, approximately

$$256^{3,840 \times 2,160 \times 3} \approx 10^{60\text{M}}. \quad (11)$$

- ▶ Furthermore, image, video, audio, and language data exhibit complex structures, so we need both high-performance and interpretable models.

Statistical Reasoning with Generative Models



- ▶ *Statistical inference* is the process of identifying the most plausible probabilistic model to describe observed data and evaluating its validity.
- ▶ Key components to overcome these challenges include:
 1. Model Space
 2. Statistical Distance
 3. Reasoning

Handwritten digit images are from Wolterink et al. (2017).

Lecture Topics

- ▶ This course covers the mathematical foundations of these pillars:

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Statistical Estimation and Hypothesis Testing
4. Basics of Probabilistic Modelling

M2. Likelihood Ratio-based Generative Models

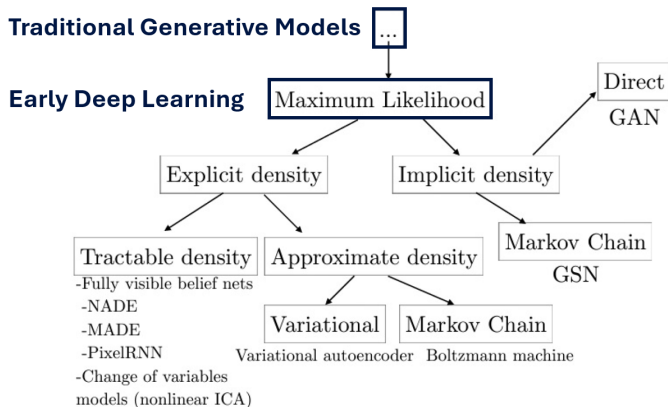
5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics

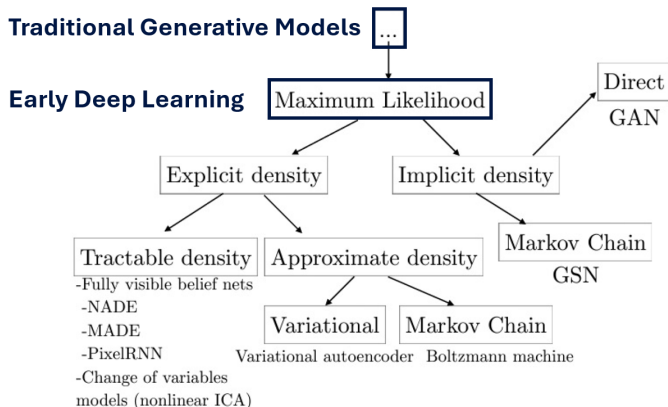
- **M1. Foundations of Probabilistic Modelling:** Core foundations for probabilistic and generative modelling (Probability theory, MLE, Bayesian networks, PCA/ICA).



The figure is edited from Goodfellow (2016).

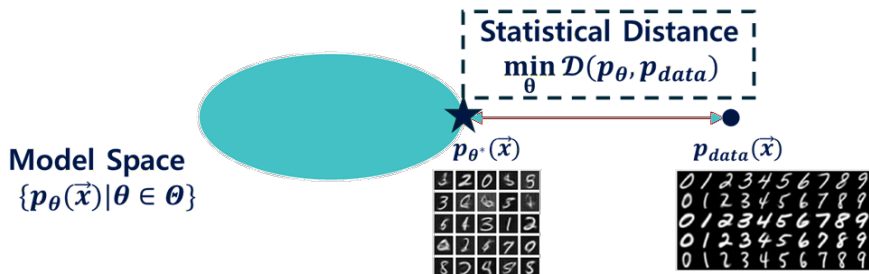
Lecture Topics

- **M2. Likelihood Ratio-based Generative Models:** Representative likelihood ratio-based models prevalent through 2014 (Autoregressive, EBM, VAE, and GAN).



Lecture Topics

- **M3. Generative Models with Other Statistical Distance:** Intuitive principles of measuring distances between distributions, and early generative models applying these specific statistical distances.



Group Presentation

- ▶ During the final lecture, students will deliver group presentations.
- ▶ Each group will select a tutorial related to generative models from a prestigious AI conference.
- ▶ Using these materials, groups will conduct a flipped-learning style presentation.
- ▶ Further details, including group composition and presentation schedules, will be announced later.

References I

- Goodfellow, I. (2016). Nips 2016 tutorial: Generative adversarial networks. *arXiv preprint arXiv:1701.00160*.
- Liu, Z., Luo, P., Wang, X., and Tang, X. (2015). Deep learning face attributes in the wild. In *Proceedings of International Conference on Computer Vision (ICCV)*.
- Park, T.-j. (2022). UNIST News [Photo Essay] Writer Kim's 'Cherry Blossom Ending'. <https://admu-intl.unist.ac.kr/unist/center/news-en.do>. Accessed: 2026-04-23.
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- Wang, M.-O., Seo, J.-C., and Lim, I.-K. (2005). Probability research of wooden die for drinking game as ethnic custom mathematics. *Journal for History of Mathematics*, 18(4):67–84.

References II

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